Two predicates: complete fixed-library audit

Observation report. The historical selector remains unresolved. Exact computation on all 11 reconstructed omissions and 10 publications across six streams.

0 exact fits

122,265 predicate pairs x16 functions =1,956,240 cases

Both robust and favorable-unknown fit sets are empty.

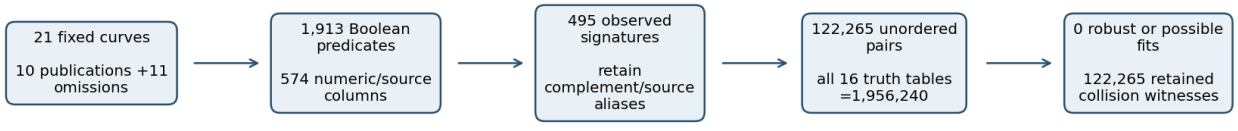
The 1,913 Boolean atoms from 574 numeric/source columns reduce to 495 complemented observation signatures. Every original alias and every pair collision witness is retained. Single-signature rules also give zero fits.

The library covers j/coefficient/discriminant power residues, local CM splitting and conductor valuations, source rational-isogeny roots at 61 levels through 293, twist observations, source ECPP flags and costs, seed expansions and four sign-loss models.

The Python integer/bitset implementation and independent PARI arithmetic/rowwise truth-table implementation agree on every feature, fit and collision witness. Embedded positive XOR, negative collision, unknown-value and A/A controls all pass.

Attempt 01 retained one completed search and one verifier stop: its wrapper rejected a benign GP stack warning. Attempt 02 corrected only that exact warning check, then completed both arms. Original logs, frozen source and failed/corrected exports are retained.

Scope: this excludes Boolean functions of two atoms in the declared library only. It does not exclude arbitrary complicated rules, field-dependent paths, undocumented stopping logic or another historical implementation. Labels and source observations were already visible; no statistical or historical attribution claim.



Python integer arithmetic + bitset bins / independent PARI arithmetic + rowwise truth tables
Both implementations agree on every feature, fit and collision witness.

This finite two-atom library only; historical selector remains unresolved.

Curve observations across all six streams

Fixed-cohort arithmetic and source-root observations

	j is a cube Boolean	rational roots ell=3	rational roots ell=5	rational roots ell=13	v3(Delta_pi) valuation	v5(Delta_pi) valuation
omission 863711343 / 36	1	0	0	2	0	0
omission 863711343 / 1562	1	0	1	2	0	1
publication prime239v1 / 2185	1	0	0	0	0	0
publication prime239v2 / 2320	1	0	1	0	0	2
omission 863711343 / 2491	1	0	0	1	0	0
omission 863711343 / 2682	1	0	1	1	0	3
omission 863711343 / 2863	1	0	1	2	0	1
publication prime239v3 / 3048	1	0	1	0	0	1
publication prime192v1 / 139	1	0	1	2	0	1
omission 863966734 / 2198	1	0	0	0	0	0
publication prime192v3 / 5228	1	0	1	2	0	1
omission 863966734 / 5405	1	0	0	2	0	0
publication prime192v2 / 6013	1	0	1	0	0	1
publication secp521r1 / 1136	1	0	1	2	0	1
omission 871696354 / 876	0	1	0	0	1	0
publication prime256v1 / 1842	0	1	1	2	1	1
publication secp384r1 / 2688	1	0	0	0	0	0
omission 891607927 / 777	0	1	1	0	1	1
omission 891607927 / 1485	0	1	0	0	1	0
omission 891607927 / 5736	0	1	0	1	1	0
publication secp224r1 / 7026	1	4	0	0	3	0

NIST-TWO-RULE-01 v2; 21 exact curves. Printed values are exact; color is scaled within each column.
Blue row labels:10 publications; orange:11 reconstructed omissions. No statistical uncertainty or historical attribution.

P-224: j is non-cubic on all three omissions and cubic on the publication. No second library atom extends this pattern to an exact six-stream classifier.

CM distinctions, custody and replay

P-256: both the published and omitted curves have exact Frobenius conductor f_pi=1, forcing actual conductor f_E=1. The published P-224 curve has f_pi=3. Ramification of 3 in D_K is different from conductor divisibility.

Local CM-field characters are obtained exactly without full global factorization: for odd ell write Delta_pi=ell^v*u. Odd v gives ramification; even v gives the Legendre symbol of u. Odd traces give (D_K/2)=(Delta_pi/2). No full P384 endomorphism order or transfer advantage is inferred.

All 21 j-1728 values are fourth powers. P-256 publication and omission both have non-cubic j. Root/coefficient sign choices do not supply the missing class exclusion. These are observations, not historical acceptance requirements.

Custody: 70 sealed files. Archive SHA256 51036fe8c46651ee12db8e1a77f1fc5d851d7d6c9c81ad81086c8735ac7b7561. Source imports are bound to exact UIDs, models and seeds; all 1,281 fibers match p and j. Original third-party source packages are not redistributed.

Frozen source commits: 8a02d2e055251b8fd0c6c142c7433bd57993b912 and b9816392a140d245b9079de67b05c392e7d39da0. Protocol v2 SHA256: 2797d0041be4fc65913b2ecd809f148ff2a87fe64a318f4cc245744bc72bd60f. Both exports pass full file verification.

Replay: extract evidence.tar.zst into a fresh directory, enter evidence/attempt-02, then run python3 -B run.py with Python3.14 and /opt/homebrew/bin/gp. A complete decompression/relocation replay passed. Replay counts are validation, not new independent cohort evidence.

Canonical artifact delivery returned HeadObject403; exact bytes are retained in Git under the existing PR2602 exception. PR merge conflicts remain unresolved. No object-store or CI success is asserted.

Primary sources retrieved 2026-10-27: words.filippo.io/seeds-solved/ and anthropic.com/research/ecdsa-curve-origins. Internal source bindings and limitations are in inputs.json and LIBRARY.md; full evidence and editable figures accompany report.md.

Next discriminating evidence: an authenticated dated caller or validation requirement, or a separately frozen field-dependent source mechanism covering every publication and omission. More elaborate label-fitted classifiers alone cannot establish historical use.